LECTURE 05 • STUDY NOTES

Input & Output Devices

Keyboard, Mouse, Touchscreen · Monitors (LCD/LED) · Printers (Inkjet/Laser)

BSc Computer Science (First Year) · Detailed notes with diagrams · 2-mark & 14-mark questions

Learning objectives. After studying these notes you should be able to:

- Classify a device as input or output and explain why.
- Explain how a keyboard and a mouse turn actions into data.
- Describe how a touchscreen senses a touch (resistive vs capacitive).
- Compare LCD and LED monitor technology.
- Compare inkjet and laser printing.

1. Input vs output devices

Input devices accept data from the outside world and convert it into binary form for the computer to process. **Output devices** take the computer's binary results and convert them into a form people can see, hear, or otherwise sense.

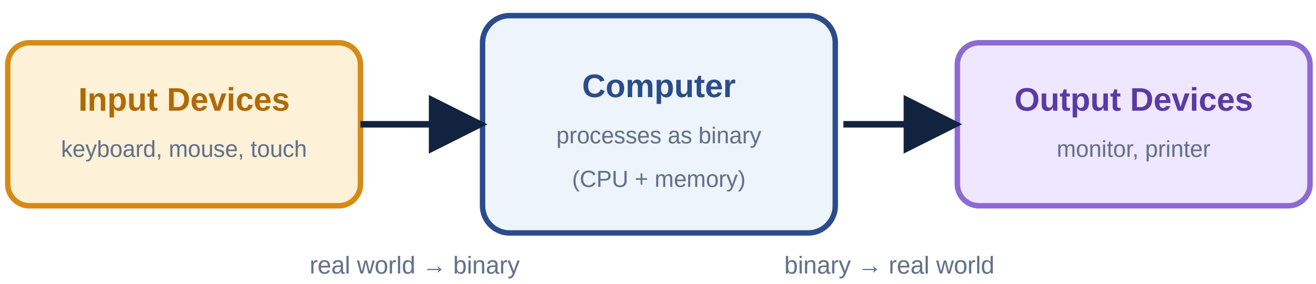


Fig 1. Input devices feed the computer; output devices present its results.

Some devices do both at once — a **touchscreen** senses touch (input) while displaying the picture (output) on the very same surface.

2. The keyboard

A keyboard is an input device built from a grid of switches, one under each key — the same basic idea as the transistor switch. Pressing a key closes its switch; a small controller inside the keyboard detects this and reports it.

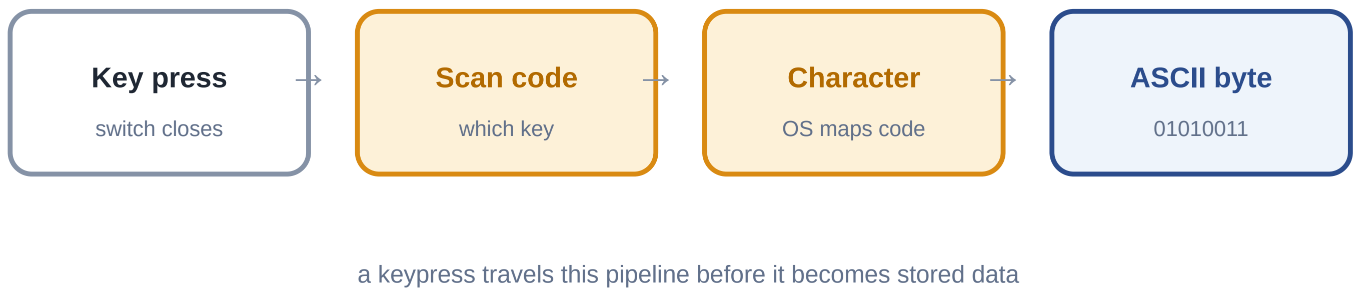


Fig 2. From a physical keypress to a stored ASCII byte.

- **Scan code** — a number identifying *which key* was pressed (or released), sent by the keyboard's controller.
- **Character mapping** — the operating system converts the scan code into the correct character, taking into account Shift, Caps Lock, and the keyboard layout.
- **ASCII/binary storage** — the character is finally stored using a code such as ASCII (see Lecture 01), e.g. 'S' = 83 = `01010011`.

3. The mouse

A mouse is a **pointing device**. Older mechanical mice used a rolling ball; almost all mice today are **optical mice**, which use a tiny LED and image sensor underneath to repeatedly photograph the surface and compare successive images to work out how far, and in which direction, the mouse has moved.

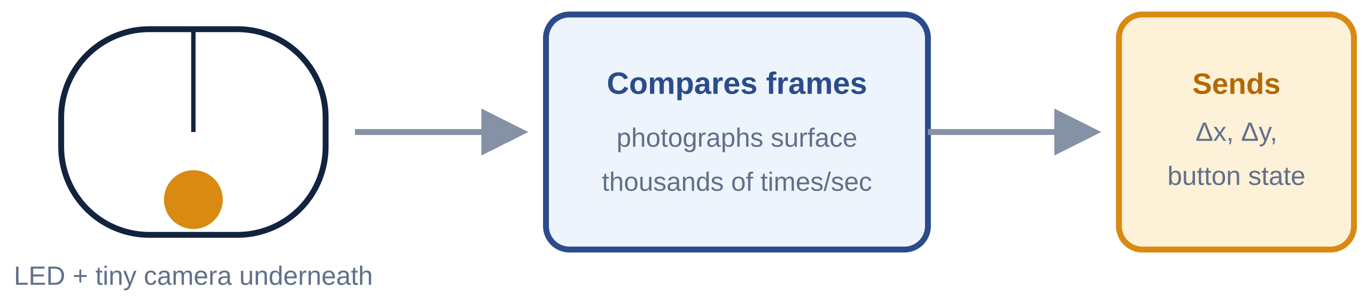


Fig 3. An optical mouse compares surface images to compute movement, then reports it as data.

- The mouse does **not** send a picture of the desk — only the **change in position** (often written Δx, Δy) and the state of its buttons (pressed or released).
- The operating system uses this stream of small movements to move the on-screen pointer and detect clicks, double-clicks and drags.

4. The touchscreen

Definition. A touchscreen is a display that can also detect and locate a touch on its surface, combining an input device and an output device in one component.

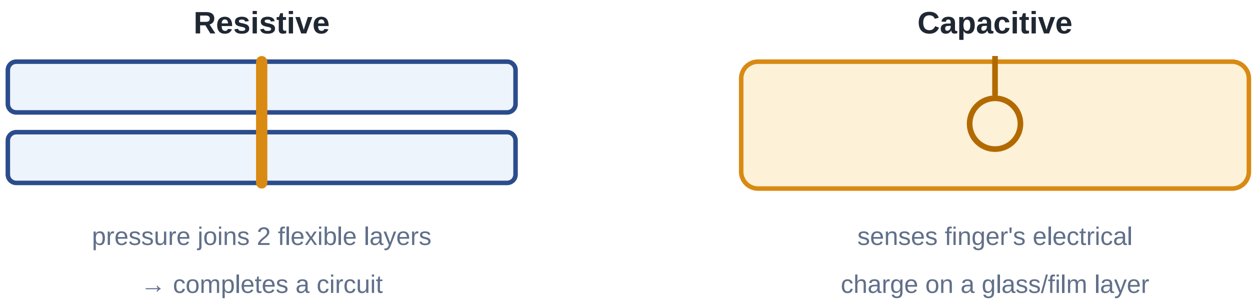


Fig 4. Resistive screens sense pressure between two layers; capacitive screens sense the finger's electrical charge.

Technology	How it senses touch	Traits
Resistive	Two flexible layers are pressed together at the touch point, completing an electrical circuit	Works with any object (finger, gloved hand, stylus); needs firmer pressure; less crisp display; cheaper to make.
Capacitive	Detects the electrical charge carried by a bare fingertip touching a conductive glass/film layer	Very responsive to light touch; supports multi-touch gestures (pinch/zoom); does not respond well to gloves or a plain stylus; used in almost all smartphones.

5. Monitors: LCD and LED

A monitor is an **output device** that displays the computer's results visually.

LCD (Liquid Crystal Display). Works by twisting liquid crystals to block or let through light from a **backlight**; colour filters give each pixel its red, green or blue. On its own the liquid-crystal layer does not produce light — it only controls how much of the backlight's light passes through.

LED monitor. Not a different display technology — it is an **LCD panel that uses LEDs (Light-Emitting Diodes) as its backlight**, instead of the older CCFL (fluorescent tube) backlight. This makes the screen slimmer, cooler, more energy-efficient, and often gives better contrast.

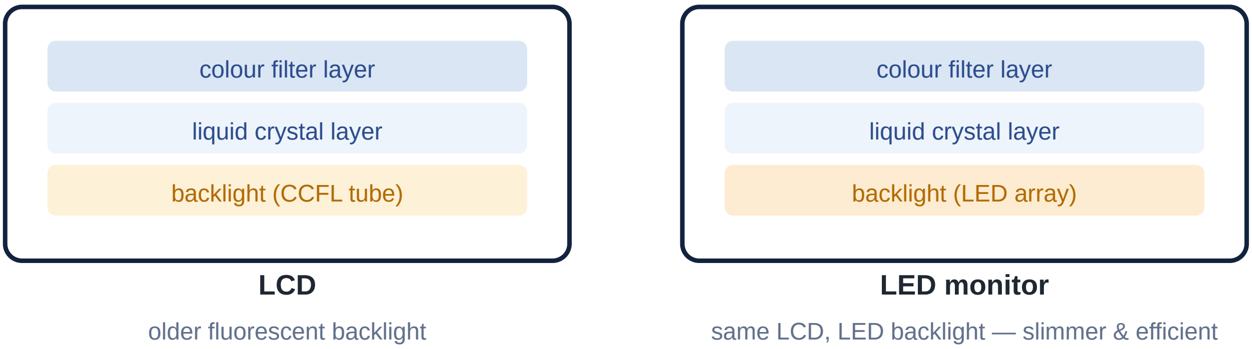


Fig 5. Both LCD and 'LED monitors' use the same liquid-crystal and colour-filter layers — only the backlight technology differs.

	LCD (CCFL backlight)	LED monitor (LED backlight)
Backlight type	Fluorescent (CCFL) tubes	Light-Emitting Diodes
Thickness	Thicker	Slimmer
Energy use	Higher	Lower
Contrast/brightness	Good	Usually better

6. Printers: inkjet and laser

Printers are output devices that convert the computer's data into a permanent, physical copy on paper — hard copy.

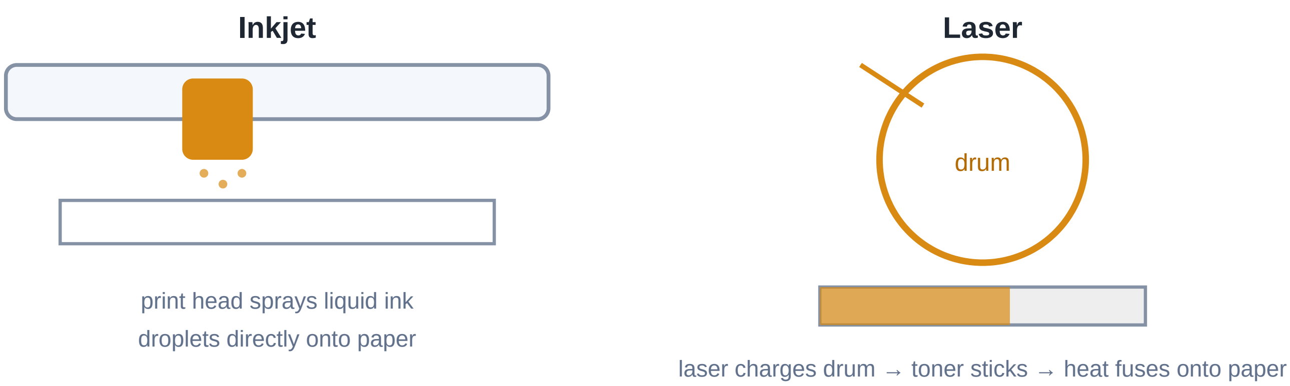


Fig 6. Inkjet sprays liquid ink; laser uses charge, light and heat to fuse toner powder.

- **Inkjet printer.** A moving print head sprays thousands of microscopic droplets of liquid ink (typically cyan, magenta, yellow and black) directly onto the paper to build up text and images.
- **Laser printer.** A laser beam draws the page as a pattern of electrical charge onto a rotating drum. Charged powder called **toner** sticks only to the charged areas of the drum, which then presses the toner onto the

paper; heated rollers **fuse** the toner permanently onto the page.

Feature	Inkjet	Laser
Method	Sprays liquid ink droplets	Laser + charged drum + toner + heat fusing
Speed	Slower	Fast, especially for text
Cost per page	Higher (ink cartridges)	Lower for high volumes
Best for	Photos, colour-rich documents, home use	Bulk text documents, offices
Output quality	Excellent for photos	Crisp, sharp text

Chapter summary. Input devices (keyboard, mouse, touchscreen) convert human actions into binary; output devices (monitors, printers) convert binary back into something we can see or hold. A keyboard reports scan codes that become ASCII bytes; a mouse reports relative movement and button state; a touchscreen combines input and output; an LED monitor is simply an LCD with an LED backlight; and inkjet and laser printers use entirely different physical processes to put ink on paper.

2-Mark Questions (Short Answer)

Answer in 2–4 lines. These test definitions and single facts.

Q1. Differentiate between input and output devices. 2 marks

Input devices accept data from the outside world and convert it into binary for the computer to process (e.g. keyboard). Output devices convert the computer's binary results into a form people can sense (e.g. monitor).

Q2. What is a scan code? 2 marks

A scan code is a number sent by the keyboard's controller identifying exactly which key was pressed or released, before it is mapped to a character.

Q3. How does an optical mouse detect movement? 2 marks

An optical mouse uses a small LED and an image sensor to repeatedly photograph the surface beneath it and compare successive images, calculating how far and in which direction it has moved.

Q4. What data does a mouse send to the computer? 2 marks

A mouse sends its relative change in position (Δx, Δy) and the state of its buttons (pressed or released); it does not send an image of the surface.

Q5. Why is a touchscreen considered both an input and an output device? 2 marks

It is an output device because it displays the picture, and an input device because it senses and locates a user's touch on the very same surface.

Q6. Differentiate between resistive and capacitive touchscreens. 2 marks

A resistive screen senses touch through pressure that joins two flexible layers into a circuit. A capacitive screen senses the electrical charge of a bare finger and supports multi-touch.

Q7. What is an LCD monitor? 2 marks

An LCD (Liquid Crystal Display) monitor uses liquid crystals to block or pass light from a backlight, with colour filters giving each pixel its colour.

Q8. Is an LED monitor a completely different technology from an LCD monitor? 2 marks

No — an LED monitor is an LCD monitor that uses LEDs instead of CCFL fluorescent tubes as its backlight; the liquid-crystal and colour-filter layers are the same.

Q9. What is the basic working principle of an inkjet printer? 2 marks

An inkjet printer's moving print head sprays microscopic droplets of liquid ink directly onto the paper to form text and images.

Q10. What is the basic working principle of a laser printer? 2 marks

A laser draws the page as charge on a rotating drum; charged toner powder sticks to that pattern and is then pressed onto the paper and fused permanently by heat.

Q11. Give one advantage of a laser printer over an inkjet printer. 2 marks

A laser printer is generally much faster and cheaper per page for high-volume text printing.

Q12. Give one advantage of an inkjet printer over a laser printer. 2 marks

An inkjet printer generally produces better quality, more vivid results for photographs and colour-rich images.

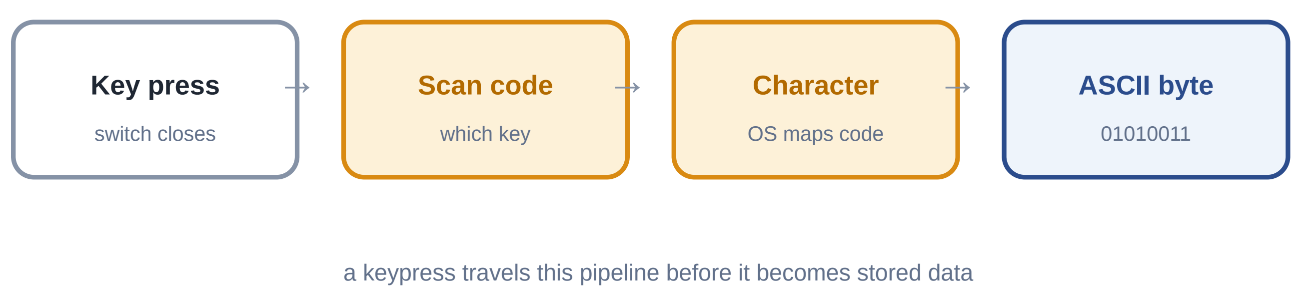
14-Mark Questions (Long Answer)

Long answers should include a definition, well-organised points, at least one labelled diagram, and a short conclusion.

Q1. Explain, with the help of a diagram, how a keystroke on a keyboard and a movement of a mouse are converted into data the computer can use. 14 marks

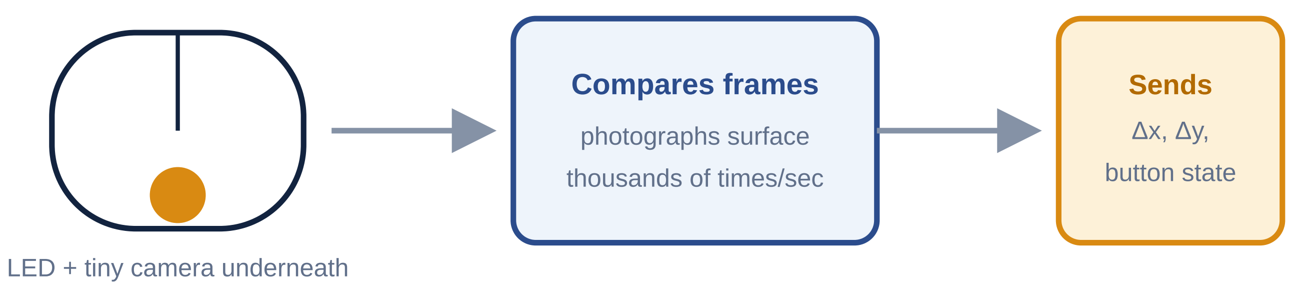
Introduction. Input devices exist to translate human actions — pressing a key, moving a mouse — into the binary data a computer can process. Two of the most common input devices, the keyboard and the mouse, do this in very different ways.

The keyboard. A keyboard is built from a grid of switches, one beneath each key — the same basic idea as the transistor switch studied earlier in the course. When a key is pressed, its switch closes, and a small controller built into the keyboard detects this and sends out a **scan code**, a number identifying exactly which key was pressed (or released). The operating system then maps this scan code to the correct **character**, taking into account modifier keys such as Shift or Caps Lock. Finally, that character is represented and stored using a standard code such as **ASCII**; for example, 'S' becomes the number 83, stored as the binary pattern 01010011.



From a keypress to a stored ASCII byte.

The mouse. Most modern mice are **optical mice**. A small LED illuminates the surface beneath the mouse, and a tiny image sensor photographs that surface thousands of times per second. By comparing each new image with the previous one, the mouse calculates how far, and in which direction, it has moved. Rather than sending a picture, the mouse sends only the **change in position** (Δx, Δy) and the current state of its buttons to the computer.



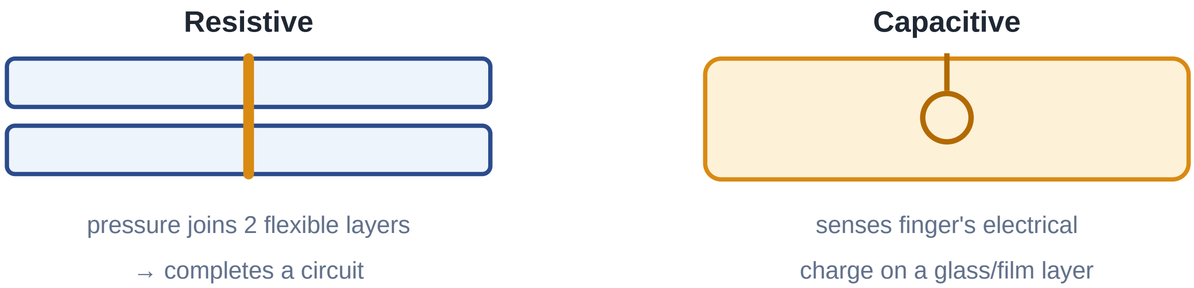
An optical mouse compares surface images to compute movement.

Conclusion. Though very different in mechanism, both devices follow the same underlying pattern of input: a physical action is sensed, converted into a small, well-defined piece of data, and passed on to the computer to be interpreted and used.

Q2. What is a touchscreen? Explain the working of resistive and capacitive touchscreens, and compare them. 14 marks

Definition. A touchscreen is a display that can also sense and locate a touch on its surface, making it both an input device (it senses touch) and an output device (it shows the picture) at the same time.

Resistive touchscreens. A resistive screen is built from two thin, flexible, electrically conductive layers separated by a small gap. When a user presses on the screen — with a finger, a gloved hand, or a stylus — the two layers are pushed together at that point, completing an electrical circuit which the device uses to calculate the exact coordinates of the touch.



Resistive senses pressure between two layers; capacitive senses electrical charge.

Capacitive touchscreens. A capacitive screen is coated with a material that stores an electrical charge. A bare finger touching the glass draws a tiny amount of this charge away at that point, which sensors detect and use to calculate the touch location. Because it relies on the finger's natural conductivity, a capacitive screen usually will not respond to gloves or a plain plastic stylus, but it is highly sensitive to light touches and can track several fingers at once, enabling gestures such as pinch-to-zoom.

Comparison.

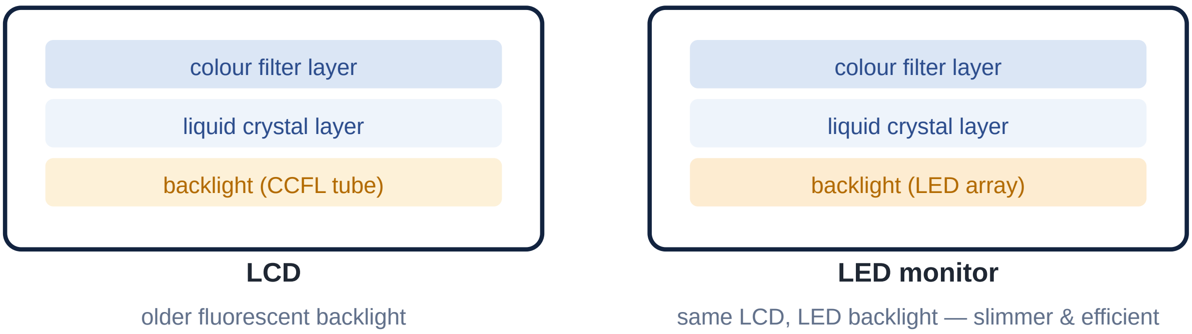
Feature	Resistive	Capacitive
Activated by	Any pressure (finger, glove, stylus)	A bare finger's electrical charge
Touch needed	Firmer press	Light touch
Multi-touch	Usually not supported	Supported
Display quality	Slightly reduced	Better clarity
Typical use	Older devices, industrial equipment	Smartphones, tablets, modern laptops

Conclusion. Both technologies achieve the same goal — locating a touch — but by sensing different physical properties: pressure in a resistive screen, and electrical charge in a capacitive one, which is why capacitive screens dominate modern touch devices.

Q3. Explain the working of an LCD monitor. What is an LED monitor, and how does it differ from a standard LCD? 14 marks

Introduction. The monitor is the most common output device, converting the computer's binary results into a visible picture. Most modern monitors are built on LCD technology.

How an LCD works. An LCD (Liquid Crystal Display) panel is made of several layers. A layer of liquid crystals can be electrically controlled to twist and either block or allow light to pass through at each point on the screen. A layer of red, green and blue colour filters then gives each of these points — each **pixel** — its final colour. Crucially, the liquid-crystal layer does *not* produce any light of its own; it only controls how much of a separate light source, called the **backlight**, is allowed through. Without a backlight, an LCD panel shows nothing at all, no matter what pattern the liquid crystals are set to.



Both panels share the same liquid-crystal and colour-filter layers; only the backlight differs.

What an LED monitor is. An 'LED monitor' is not a separate display technology from LCD — it *is* an LCD panel, but one that uses **LEDs (Light-Emitting Diodes)** as its backlight, in place of the older CCFL (Cold Cathode Fluorescent Lamp) tubes used in earlier LCD monitors. Because LEDs are smaller, cooler and more efficient than CCFL tubes, LED-backlit monitors are typically slimmer, use less power, run cooler, and often achieve better contrast and brightness.

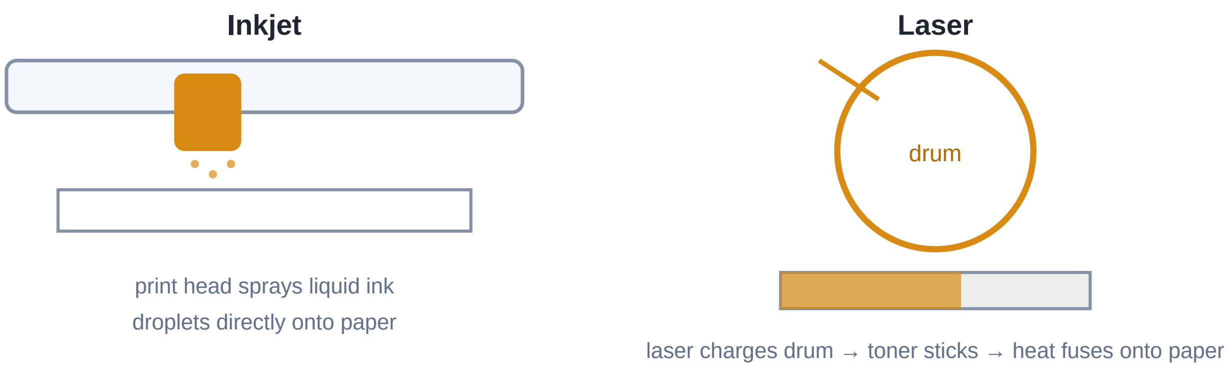
Comparison.

	LCD (CCFL backlight)	LED monitor (LED backlight)
Backlight	Fluorescent tubes	Light-Emitting Diodes
Thickness	Thicker	Slimmer
Power use	Higher	Lower

Conclusion. Understanding that 'LED monitor' describes the backlight, not a wholly new display type, clarifies a common point of confusion: every LED monitor is an LCD monitor, simply built with a more modern, efficient light source behind the same liquid-crystal picture layer.

Q4. Compare inkjet and laser printers in terms of their working principle, speed, cost and best use, with the help of diagrams. 14 marks

Introduction. A printer is an output device that produces a permanent, physical copy — a hard copy — of the computer's data on paper. The two most common types, inkjet and laser, use entirely different physical processes to achieve this.



Inkjet sprays liquid ink; laser uses charge, toner and heat.

Inkjet printers. An inkjet printer's print head moves back and forth across the page. As it moves, tiny nozzles spray thousands of microscopic droplets of liquid ink — typically cyan, magenta, yellow and black — directly onto the paper, building up text and images line by line. Inkjet printers are well suited to producing vivid, detailed colour images such as photographs, and are common in homes due to their lower upfront cost, though the ink cartridges themselves can be relatively expensive over time.

Laser printers. A laser printer works very differently. A laser beam is used to draw the page, one line at a time, as a pattern of electrical charge on a rotating drum. A fine powder called **toner** is then applied to the drum; it clings only to the areas that carry the laser's charge pattern. The drum presses this toner pattern onto the paper, and the page then passes through heated rollers that **fuse** the toner permanently onto the paper. Laser printers are significantly faster, especially for text, and have a lower cost per page at high volumes, making them the standard choice for offices.

Comparison table.

Feature	Inkjet	Laser
Method	Sprays liquid ink droplets	Laser + charged drum + toner + heat
Speed	Slower	Fast, especially for text
Cost per page	Higher	Lower at high volume
Best for	Photos, colour documents, home use	Bulk text, office use

Conclusion. Inkjet and laser printers trade off differently between quality, speed and cost: inkjet excels at rich colour images for occasional home use, while laser excels at fast, economical, high-volume text printing, which is why it dominates office environments.